

Manav Ananthakumar

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Education

Arizona State University: MS in Data Science, Analytics and Engineering

Aug 2025 – May 2027

Relevant Courses: Machine Learning, Big Data Analytics, Data Mining (GPA 3.89/4.0)

Tempe, AZ

Manipal Institute of Technology: B.Tech, Computer Science and Engineering

Jul 2019 – Jul 2023

Relevant Courses: Machine Learning, Operating Systems, Computer Networks (GPA 3.5/4.0)

Manipal, India

Work Experience

Shell

Aug 2023 – Aug 2025

Associate Data Engineer

Bangalore, India

- Owned end-to-end **SAC OSE** migration for Shell's enterprise dashboards, re-engineering **7+ legacy stories** connected to multi-instance **SAP ECC** sources and validating data parity across business units post-cutover.
- Revamped **SAP HANA** calculation views across Downstream and Group Finance targets (P94, GF HANA), cutting data processing latency by **15%** on enterprise-wide financial reporting pipelines.
- Built a **Python** automation framework for the **Nucleus brownfield ECC-to-S/4HANA** migration, applying surrogate key logic to cross-compare data dumps across **8+ ECC instances** and surface discrepancies ahead of each TM cutover.
- Validated **S/4HANA-to-SAC** data integrity across Downstream, Group Finance, and Upstream HANA targets post-migration, serving as the sign-off layer for production go-live readiness and ensuring a seamless enterprise-wide transition.

Bosch Limited

Jan 2023 – Jun 2023

Data Science Intern

Bangalore, India

- Built an **LSP recommendation system** in **Python** and **Streamlit** over **13K+ trip records** across 201 routes; invented a **distance-slab bucketing** approach (250km intervals) ranking providers by cost/kg and delay score, cutting delivery times by **12%**.
- Designed an **NLP pipeline** on 3 years of GPTW survey data using **VADER** sentiment scoring, **text2emotion** classification, and **NLTK n-gram** analysis; delivered networkx co-occurrence graphs and word clouds to HR, improving employee satisfaction by **8%**.
- Traced data quality issues in **Travis 2.0** logistics records (incorrect dispatch dates, invalid pincodes) via **Power BI** profiling and Python validation, improving **data completeness by 15%** and **forecast accuracy by 7%**.

Deloitte USI

May 2022 – Jul 2022

Tax Technology Intern

Bangalore, India

- Built a **Document Management System** in **ASP.NET Core 8 MVC + EF Core + SQLite** with bcrypt-authenticated login, role-scoped file listing, and **.pbix-only upload validation** — enforcing Deloitte's Power BI file policy across **4 screens**.
- Modelled a **6-sheet star schema** (~1M rows, 14 manufacturers) in **Power BI**, authoring **DAX measures** for rolling-12M market share, YTD units, and YoY delta to deliver a **3-page Nike sales dashboard** adopted by **10+ stakeholders**.

Relevant Projects

PitchMirror – *React 19, Node.js, Gemini 2.5 Flash Native Audio, AssemblyAI, LangChain, Firecrawl, Anam SDK*

- Engineered a multi-agent AI interview simulation platform using **Gemini 2.5 Flash Native Audio**, **AssemblyAI** transcription, and a **LangChain ReAct** agent loop with **Firecrawl** for company-aware pre-session research — **3rd at MLH Innovation Hacks 2.0** (80 teams).
- Implemented role-specific evaluation pipelines (recruiter, investor, professor, coding) and thread-level progress tracking with hidden steering memory that personalises future sessions without breaking simulation realism.

QuantaMine – *FastAPI, Next.js 15, Gemini API, FinBERT, NewsAPI, SSE, Docker*

- Developed an AI-driven platform using **Python** and **Gemini API** to generate, backtest, and evaluate algorithmic trading strategies, translating natural-language strategy descriptions into executable **Python** models via LLM code generation.
- Built a model evaluation engine computing **Sharpe ratio**, **max drawdown**, and **cumulative return** metrics, enabling systematic performance comparison across generated strategy variants against benchmark indices.

RegLens – *FastAPI, LangChain, OpenAI, Qdrant, Elasticsearch, Docker*

- Built a **RAG** Compliance Intelligence Platform using **Python**, **FastAPI**, **LangChain**, **OpenAI embeddings**, and **Qdrant** for citation-backed retrieval over legacy financial policy documents with metadata-aware versioning.
- Implemented **hybrid search (vector similarity + BM25)** via Elasticsearch with **cross-encoder reranking**, improving retrieval precision and reducing hallucinated responses on complex regulatory queries.

Technical Skills

Programming: Python, C/C++, SQL, React, ASP.NET Core

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers

AI & LLM Engineering: RAG, Prompt Engineering, LLM Evaluation, Hybrid Retrieval, Embeddings, Cross-Encoder Reranking

LLM APIs & Frameworks: OpenAI API, Gemini API, Claude API, LangChain

Data Platforms: Databricks, Microsoft Azure, Qdrant, Elasticsearch (BM25), SAP HANA, SAP S/4HANA

Backend & DevOps: FastAPI, Docker, MLflow, Linux

Visualization: Streamlit, Tableau, Microsoft Power BI, SAP Analytics Cloud